

Introduction to HTML and CSS - Exercises

Topics Covered

- What is HTML?
- HTML Syntax
- Tags:
 - html, head, title, meta, body,
 - h1, h2, h3, h4, h5, h6, p, a, strong, em, ul, ol, li, dl, dt, dd, img, blockquote
 - main, aside, article, section, header, footer
 - details, summary, figure, caption
- What is CSS?
- CSS Syntax
- Properties
 - background, color
 - border, border-radius
 - padding
 - font-family, font-weight, font-style
 - text-decoration
 - list-style

Resources

- HTML Element references on MDN:
 - <https://developer.mozilla.org/en-US/docs/Web/HTML/Element>
- HTML Basics
 - https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web/HTML_basics
- Semantics:
 - <https://developer.mozilla.org/en-US/docs/Glossary/Semantics>
- W3 Validator:
 - <https://validator.w3.org>
- CSS Basics:
 - https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web/CSS_basics
- CSS Reference:
 - <https://developer.mozilla.org/en-US/docs/Web/CSS/Reference>

Tasks

1. Simple image gallery

Build a simple image gallery using HTML and CSS to display a collection of images.



Requirements

- **HTML Document Setup:**
 - Create an **HTML** document named **image-gallery.html**.
 - Include a **<div>** container to hold the image gallery.
- **CSS File Setup:**
 - Create a **CSS** file named **gallery-styles.css**.
 - Link the **gallery-styles.css** file to your **image-gallery.html** document using the **<link>** tag in the **<head>** section.
- **Gallery Structure:**
 - Inside the gallery container, use **** tags to include the 12 provided images.
 - Optionally, wrap each **** tag in an **<a>** tag so that the images can be clicked to view the full size in a new tab.

Resources

- <https://unsplash.com/>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/img>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/div>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/ul>

2. ToDo List

Build a simple ToDo List using HTML and CSS.

Stuff to do

1. Trash
2. Dishes
3. Vacuum

Done

1. Eat-chocolate
2. Eat-pizza

Requirements

- **HTML Document Setup:**
 - Create an **HTML** document named **todo-list.html**.
 - Include a basic **HTML** structure (**<!DOCTYPE html>**, **<html>**, **<head>**, **<body>**).
- **CSS File Setup:**
 - Create a **CSS** file named **todo-list.css**.
 - Link the **todo-list.css** file to your **todo-list.html** document using the **<link>** tag in the **<head>** section.
- **Structure:**

- Use 2 unordered lists to represent tasks that are outstanding and tasks that are done

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/ul>
- <https://developer.mozilla.org/en-US/docs/Web/CSS/text-decoration>

3. Navigation Bar

Build a simple navigation bar using HTML and CSS that includes links to different sections of a website.

• [Home](#)
• [About](#)
• [Services](#)
• [Contact](#)

Requirements

- **HTML Document Setup:**
 - Create an **HTML** document named **navigation.html**.
 - Include the basic **HTML** structure (doctype, html, head, body).
- **CSS File Setup:**
 - Create a **CSS** file named **navigation.css**.
 - Link the **navigation.css** file to your **navigation.html** document using the **<link>** tag in the **<head>** section.
- **Structure:**
 - The navigation bar should be contained within a **<nav>** element.
 - Include an unordered list (****) within the **<nav>** element. Each item in the list (****) represents a link (**<a>**) to a section of the website (e.g., Home, About, Services, Contact).

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/nav>

4. List of blog posts

Develop a simple webpage that displays a list of blog posts, each with a title, an excerpt, and meta information including the post time and author.

The Evolution of Web Design

March 10, 2024 by Alex Johnson

Web design has come a long way since the early days of the internet...

Understanding CSS Grid

March 15, 2024 by Sam Lee

CSS Grid is a powerful layout system available in CSS...

Requirements

- **HTML Document Setup:**
 - Create an **HTML** document named **list-of-posts.html**.

- Ensure the document includes the basic HTML structure (doctype, html, head, body).
- **CSS File Setup:**
 - Create a **CSS** file named **list-of-posts.css**.
 - Link the **list-of-posts.css** file to your **list-of-posts.html** document using the **<link>** tag in the **<head>** section.
- **Blog Posts Structure:**
 - Use an **<article>** element for each blog post to semantically represent the content.
 - Within each **<article>**, include a **<header>** element that contains the post title (**<h2>**), post time (**<time>**), and author (**** or **<p>**).
 - Follow the header with a paragraph (**<p>**) serving as the post excerpt or summary.

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/article>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/header>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/section>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/footer>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/p>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/blockquote>

5. Single blog post page with comments

Create a single blog post page that displays the post content along with a comments section below it.

Blog Post Title

Posted by **Author Name** on January 24, 2024

This is the introduction of your blog post...

Comments

John Doe - January 25, 2024

This is a comment on the blog post.

Requirements

- **HTML Document Setup:**
 - Create an **HTML** document named **single-post.html**.
 - Include the standard HTML structure (**<!DOCTYPE html>**, **<html>**, **<head>**, **<body>**).
- **CSS File Setup:**
 - Create a **CSS** file named **single-post.css**.
 - Link the **single-post.css** file to your **single-post.html** document using the **<link>** tag in the **<head>** section.
- **Structure:**
 - Use semantic **HTML** to structure the blog post. This should include a **<header>** with the post title (**<h1>**) and metadata such as the author and publish date (**<p>**).
 - Include a **<main>** section for the blog post content, utilizing **<p>** for paragraphs, **<h2>** for subheadings, and other relevant tags as needed for the content (e.g., **** for images, **<blockquote>** for quotes).

- Below the blog post, include a **<section>** for comments. Each comment should be contained within an **<article>** tag, with a **<header>** for the commenter's name and the comment date, and a **<p>** for the comment text.

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/article>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/header>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/section>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/footer>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/p>

6. Blog layout

Create a responsive blog layout with a main content area for blog posts and a sidebar for supplementary content, using HTML and CSS.



Requirements

- **HTML Document Setup:**
 - Create an **HTML** document named **blog-layout.html**.
 - Include the standard HTML5 structure (doctype, html, head, body).
- **CSS File Setup:**
 - Create a **CSS** file named **blog-layout-styles.css**.
 - Link the **blog-layout-styles.css** file to your **blog-layout.html** document using the **<link>** tag in the **<head>** section.
- **Blog Layout Structure:**
 - Utilize semantic **HTML** tags to structure the layout, including **<header>**, **<main>**, **<article>**, **<aside>**, and **<footer>**.
 - The **<header>** should contain the blog's title or logo and a navigation menu.
 - The **<main>** area should display blog posts, each wrapped in an **<article>** tag, including titles (**<h2>**), publication dates (**<time>**), authors, summaries (**<p>**), and read more links (**<a>**).
 - The **<aside>** should serve as a sidebar, containing widgets or links such as "About Me", categories, recent posts, or social media links.
 - The **<footer>** should contain copyright information and/or links to privacy policy, terms of service, etc.